Social context, identity and prosocial behavior

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Abstract

As society becomes increasingly diverse, a key question arises: does a change in our social context — defined by the individuals we are exposed to — influence our interactions with each other? This paper studies this question using an experiment in a large-scale U.S. sample. Participants make an incentivized allocation towards either a fellow U.S. national or a foreigner, while exogenously exposed to one out of two social contexts with varying levels of diversity in nationalities. I find that facing a diverse context amplifies ingroup bias, driven by both increased allocations towards U.S. nationals and decreased allocations to foreigners relative to allocations in a homogeneous context. Evidence suggests that changes in perceptions of social proximity are a mechanism behind the effects of context on allocations. Finally, exposure to a diverse context influence political views, in a direction consistent with an increased ingroup favoritism towards U.S. nationals.

Keywords: social context, diversity, altruism, ingroup bias, social proximity

JEL classification: D91, D64

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1 Introduction

Our neighborhoods, classrooms, and workplaces are becoming increasingly diverse (Frey, 2020). Yet even amidst this surge in diversity, segregation persists, potentially causing individuals to experience mere exposure without genuine contact with diverse groups (Boustan, 2013; Hellerstein and Neumark, 2008; Williams and Emamdjomeh, 2018). This heightened exposure to diversity raises a crucial question: whether changes in our *social context*, characterized by the individuals to whom we are *exposed*, influence the way we interact with each other. For instance, would the interaction between two white Americans, in a predominantly white neighborhood, change following an influx Chinese families?

This paper uses a preregistered large-scale experiment on a U.S. sample ($N \approx 2800$) to study whether a change in a social context has a causal effect on altruism, which is an important determinant of social cohesion and cooperation in interactions (Fehr and Fischbacher, 2003; Alan et al., 2021). Studying the effects of changes on social context on behavior comes with two main empirical challenges. First, observed changes in social contexts are often intertwined with other significant economic and social shifts, making it complex to pinpoint specific causality. Second, it is often not possible to distinguish in a social context whether an individual is just merely exposed or has genuine contact with others in their society, making it challenging to identify mechanisms and to specify the direction of the effect of a change in social context. This paper addresses these concerns by isolating the mere exposure channel in an experimental design where social context is exogenously manipulated.

Participants from the U.S. make an incentivized allocation decision towards a receiver, who is either a fellow U.S. national or a foreigner, while exogenously exposed to a social context. To mimic the varying exposure to diversity in society, the decision-maker is exposed to one out of two social contexts: (i) a *homogeneous* context, where she observes only peers from a single country — e.g., she observes only U.S. nationals or only foreigners, and (ii) a *diverse* context, where she observes both fellow U.S. nationals and foreigners simultaneously. By comparing allocations across the two social contexts, I show that the social context a participant is exposed to has a substantial impact on altruism. Facing a diverse context increases ingroup favoritism towards U.S. nationals. I provide evidence of changes in perceived social proximity towards the receiver as a mechanism of the effects social context on allocations. Moreover, the increased perceived social proximity towards fellow U.S. nationals due to changes in social context

carries over to political preferences, by making U.S. participants care more about national redistribution.

The experimental design introduces a way to exogenously manipulate social context. A decision-maker is exposed to a social context, composed by two other matched individuals. In each context, the two individuals are randomly assigned the roles of a receiver and a non-receiver, where the decision-maker can decide how much to redistribute from her endowment towards the receiver only. The non-receiver is key for the manipulation of social context. A decision-maker is in a homogeneous context when the receiver and non-receiver come from the same country; and the diverse context is the case when the receiver and non-receiver come from different countries. Thus, the experiment follows a between-subject 2x2 design, where a decision-maker faces an allocation decision which is either in a homogeneous or a diverse context, and directed towards a U.S. national or a foreigner. This design allows me to study the effects of changes in the social context on allocations towards both: fellow U.S. nationals and foreigners. In a diverse context individuals allocate 12% more to fellow U.S. nationals and 15% less to foreigners, relative to the respective allocations in a homogeneous context. Notably, in homogeneous contexts allocations towards the U.S. nationals and to foreigners are the same. Taken together, these results suggest that social context is key for the emergence of ingroup bias.

To shed light on the mechanisms behind the main results, I examine the relationship between the allocation decisions and perceived social proximity towards the receiver, measured using the Inclusion of the Other in the Self (IOS) scale, used in recent work in economics (Goette and Tripodi, 2021; Bicchieri et al., 2022; Gächter et al., 2022). The analysis follows a simple formal framework, based on two premises. First, drawing from economic studies on identity, a decision-maker places greater weight on the receiver's pay-offs when the receiver's identity is perceived as closer to their own (Chen and Li, 2009; Leider et al., 2009; Robson, 2021). Second, this perception of social proximity can be influenced by the prevailing social context. Following Bordalo et al. (2016) exposure to diversity modifies the decision-maker's reference group. By contrasting the receiver's group with the new and distinct reference group, perceptions of social proximity shift in a manner that magnifies group differences, enhancing social proximity to one's ingroup. The model predicts that higher perceived social proximity to the ingroup leads to a rise in ingroup-biased allocations.

This conceptual framework predicts that diversity increases ingroup favoritism in

both social proximity and, as a consequence, in allocations relative to a homogenous society, which I test using the collected measure of perceived social proximity. My findings provide support for the first premise, as allocation decisions and perceived social proximity exhibit a strong positive correlation. Furthermore, in diverse contexts, decision-makers feel closer to the ingroup relative to homogeneous contexts, which corresponds to evidence for the second premise. Finally, the effects on allocations are partially driven by the individuals whose proximity was affected by the social context. Collectively, the evidence suggests that the effects of context on allocations are substantially driven by shifts in perceived social proximity.

I demonstrate the robustness of the results by analyzing the effects of social context on allocations and social proximity separately for each of the foreign countries used in the experiment: China and Canada. I find that the results are consistent regardless of the foreign country used, which suggests that the effects are not driven by country-specific beliefs or attitudes. Furthermore, taking advantage of the large-scale U.S. sample, I study heterogeneity in the effects of social context over allocations and perceived social proximity exploiting a set of sociodemographic characteristics. I find no heterogeneity across education, age, sex, political affiliation, or race, and the results are consistent across these dimensions. The consistency of my findings across subgroups suggest that the study identifies a general feature in human behavior.

Finally, I explore whether the increase in perceived social proximity towards fellow U.S. nationals, following a change in social context, carries over to political preferences over redistributive policies. Participants indicated, in a hypothetical scenario, whether they agree with prioritizing nationwide over local redistribution of a tax raised at a local community level, in the form of welfare payments. My findings are twofold: first, participants indicating a higher social proximity towards other anonymous U.S. nationals exhibit a higher level of agreement with prioritizing nationwide redistribution. Second, exposure to international diversity not only increases perceived social proximity towards fellow U.S. nationals, but also enhances support over prioritizing nationwide distribution. This suggests that exposure to diversity reshapes the boundaries of the ingroup from local to national, which is consistent with work that shows how changes social context reshapes ingroup boundaries over intergroup interactions (Fouka and Tabellini, 2022).

This paper contributes to our knowledge in several strands of literature. First, this paper enriches the interdisciplinary research that delves into the implications of di-

versity and context on intergroup relations (Allport, 1954; Rao, 2019; Mousa, 2020; Lowe, 2021; Nathan and Sands, 2023). While previous studies have predominantly centered on intergroup interactions, e.g. interactions of natives and foreigners; our research shows that diversity also affects intragroup interactions, e.g., how natives interact with each other. In a recent paper, Anderberg et al. (2023) study natives decisions in a trust game towards other natives and immigrants in a classroom context, using survey data in Germany. Similar to my findings, they show that classrooom levels of diversity affect ingroup bias. My paper differs from their approach, by using an experiment that allows me to additionally assess: first, how exposure to diversity affects altruism, using a non-strategic setting; second, how mere exposure to diversity affects behavior, using a controlled setting that isolates this channel; and finally, the role of perceived social proximity as a driver of the effects of social context.

This research is grounded in theoretical work from economics and psychology on how group identity is affected by contextual factors and play a role in economic decisions (Tajfel et al., 1979; Turner et al., 1987; Heidhues et al., 2020; Grossman and Helpman, 2018; Bonomi et al., 2021). I contribute to this work by providing empirical evidence of the effect of social context on a key economic behavior, which is altruism. Moreover, drawing from theoretical research on the effects of context on beliefs and perceptions (Esponda et al., 2023; Bordalo et al., 2016), I provide evidence of a novel mechanism over which social context affects altruism, which is changes in perceived social proximity.

Relatedly, this paper enriches the literature on ingroup bias in social preferences (Luttmer, 2001; Chen and Li, 2009; Charness and Chen, 2020; Shayo, 2020; Kranton et al., 2020). While extensive research has documented the existence of ingroup bias in different settings, less work has been directed on when is this bias present or what determines its strength (Hett et al., 2020). My findings highlight the importance of social context on the intensity and presence of ingroup favoritism.

Finally, my findings contribute to the literature on the effects of social context on policy views (Condra and Linardi, 2019; Hangartner et al., 2019; Steinmayr, 2020; Fouka et al., 2022; Alesina and Tabellini, 2022). Most work focuses on understanding the effects of diversity or migrants on policy views regarding the outgroup, e.g. anti-immigrant sentiment or immigration policies. My paper instead focuses of the effects of social context over policies associated to the ingroup and links the novel mechanism of changes in perceived social proximity towards the ingroup as an additional driver of political preferences.

The remainder of the paper is structured as follows. In Section 2, I propose the experimental set up that allows me to identify social context effects on willingness to help. Section 3, defines the empirical strategy and indicate the hypotheses. In Section 4 the main results are presented, as well as suggestive evidence of the mechanism and implications on policy views. Finally, in Section 5 I discuss the potential extensions of the paper and future directions.

2 Experimental Design

I first describe how social context is defined in my experimental framework. Subsequently, I explain the main task of the decision maker (DM): the allocation decision. Then, I illustrate the manipulation of social context in the experimental setting. I proceed to provide an overview of the secondary outcomes collected in the experiment. Finally, I describe the treatment conditions and summarize the sample and procedures.

Social context. In the experiment, the group membership that characterizes social context is given by nationality, where the DM is always from USA.[1] The DM can be exposed to either a U.S. national, with whom she shares the same group membership, or a foreigner, who differs with the DM in group membership. The study focuses on two types of social contexts a DM can be exposed to. First, a *homogeneous* context, where the DM is exposed to a set of peers from a single country, e.g., the DM observes only Americans or observes only foreigners. Second, a *diverse* context, where the DM is exposed to a heterogeneous set of peers, where different countries contrast with each other, e.g., the DM observes both, Americans and foreigners simultaneously. The foreign countries used in the experiment correspond to China and Canada.

Allocation decision. I measure prosocial behavior using a redistributive allocation decision. Each DM is matched with two other participants of the study (A and B) and informed about their group membership. All three, together, share a total of $50, initially distributed between the three as follows: $40 for the respondent, and $5 for each of the matched participants.[2] After providing the information about the initial distribution to the DM, one of the two matched participants is randomly selected with equal probability, who I refer to as the *receiver*. The DM is informed about the selection process. To

[1]The exact definition of the group membership would be the country were the participant of the study was sampled from. However, for simplification in exposition, I refer to it as nationality.

[2]All participants are asked to perform the task, however, they are informed that only a randomly selected subset of the participants will receive the total amount.

determine the final pay-offs, the DM must decide how much of her initial endowment of $40 would she allocate *only* towards the receiver. Note that final pay-offs of the DM and the receiver depend on the allocation choice of the DM, whereas the final pay-off of the unselected participant, which I refer to as the *non-receiver*, is $5 regardless of the choice of the DM. Nonetheless, the presence of the non-receiver is key for manipulating social context.

Importantly, the respondent is informed that she is the only participant in their group that is able to change the initial distribution of the pay-offs, and that the decision is anonymous. This is done as a means to withdraw potential signaling and social image concerns in determining the allocation decision (Andreoni and Bernheim, 2009).

Figure 1 shows a screen capture of two potential situations a DM may be subject to. In each panel, the DM is matched with two participants, and is informed about the country of each participant. In red, it is indicated which participant was randomly selected as a receiver. In the first situation (Panel 1a), the DM (USA) is matched to participant A and B that are both fellow U.S. nationals. Participant B was selected, and therefore the DM must decide how much to allocate towards a fellow U.S. national. In the second situation (Panel 1b), the DM is matched to a participant A that is a foreigner and a participant B that is a fellow U.S. national. Participant B was selected, and therefore the DM must decide on how much to allocate towards a fellow U.S national. This example highlights the two key features of the design. First, in both of the situations the decision and choice set is the same, corresponding to an allocation towards a U.S. national. Second, across situations the only aspect that differs is social context: the first situation corresponds to a homogeneous context (A and B are both U.S. nationals), whereas the second situation corresponds to a diverse context (A is U.S. national and B is a foreigner). Note that social context is manipulated by only modifying the non-receiver.

Social proximity elicitation. To understand the effects of social context on perceptions in the social identity domain, I elicit a proxy measure of social proximity between individuals. Using the Inclusion of Other in the Self (IOS) scale developed by Aron et al. (1992), I ask the respondents to indicate *how close do they feel* towards the receiver using two overlapping circles, where no overlap indicates not close at all, and the bigger the overlap between the circles, the closer the respondent feels. This measure is unincentivized, however, it has been validated as a reliable measure for social proximity in comparison to other more sophisticated survey methods (Gächter et al., 2015) and have seen an increased use in recent experiments in psychology and economics (Goette and

Figure 1: Example of a change in social context from homogeneous (above) to diverse (below) in the allocation toward U.S. nationals.

(a) Treatment **Homogeneous - U.S. National**

(b) Treatment: **Diverse - U.S. National**

Tripodi, 2021; Bicchieri et al., 2022; Gächter et al., 2022).[3]

Policy views. Based on the work of Enke et al. (2022) and Cappelen et al. (2023), I elicit preferences over policy views, where respondents indicate their support towards a redistributive policy. The participants are asked to indicate their agreement in a 5 points scale from "Strongly Disagree" to "Strongly Agree" (-2 to 2), towards the following policy:

"The government should redistribute local tax revenues as welfare payments across all communities nationwide, rather than only within the local communities they were raised."

Other outcomes. In the survey, key demographic information such as education,

[3]The order of presentation of the allocation task and the social proximity elicitation is randomized.

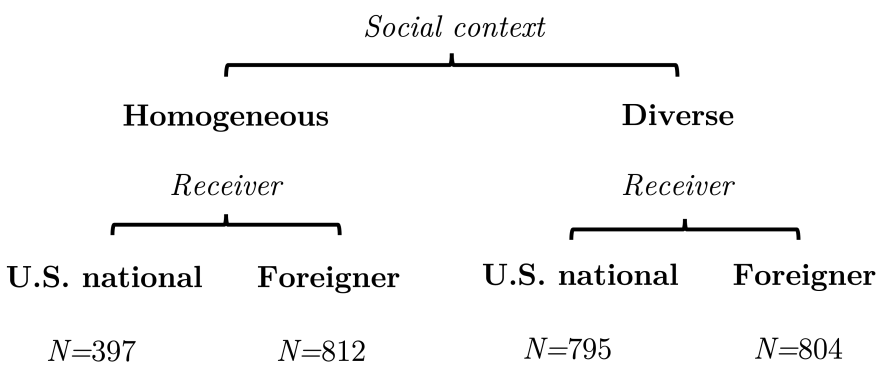
political affiliation, age, ethnicity, and gender are collected.

2.1 Treatment conditions

Figure 2 provides an overview of the experimental design. I use a 2x2 between-subject design, where subjects are randomized across two dimensions. First, whether the receiver is a U.S. national or a foreigner, from either China or Canada. Second, whether the allocation takes place in a homogeneous context, where both matched participants come from the same country (e.g., A and B are both form China) or a diverse context, where one matched participant is from the U.S., and the other is from a foreign country.[4] The randomization procedure defines the following four treatments:

1. Homogenous - U.S. National. $N \approx 400$.
2. Homogenous - Foreigner. $N \approx 800$.
3. Diverse - U.S. National. $N \approx 800$.
4. Diverse - Foreigner. $N \approx 800$.

Figure 2: Overview of the randomization in the experiment



In the latter treatments that involve the foreigners, I split the subsamples according to which country was used as the foreign country: China or Canada. The decision of selecting China and Canada as the foreigners emerged from the interest in analyzing

[4]This setting considers societies with members only from USA and/or Canada, or societies with members only from USA and/or China. Other situations that combine both of the foreigners together (Canada and China) are outside the scope of this paper.

the context effects from two different types of outgroups, defined in the preregistration. First, the *strong outgroup* (China) which corresponds to a foreign group that is dissimilar in observable characteristics to the USA group. Second, I define the *weak outgroup* (Canada), which corresponds to a foreign group that is more similar to the USA group. I expected, accordingly, that the social proximity towards participants from China and Canada would be different. Nonetheless, Figure 8 in Appendix A shows that there is no difference in social proximity towards the two selected groups of foreigners, in the homogeneous setting. I interpret this finding as if both Canada and China are perceived as an equal outgroup. As a result, I focus on the analysis of the pooled sample, using both countries together as foreigners.

2.2 Experimental procedures

The survey was conducted online by survey provider Dynata. The company recruited a representative sample of the US population, between October 14th-31st 2022. There was a total of 2808 valid participants who passed an initial mandatory attention check. Respondents were stratified to match adult population by age, sex and geography. The average response duration was of around 5 minutes. The experimental design and the hypotheses were preregistered at the AEA RCT Registry (AEARCTR-0010179). The full instructions are made available in Appendix D.

Table 4 in Appendix A provides a summary of the outcomes collected in the experiment. The sample consists of around 60% of women, around a third identifying as Republican, a mean age of 50, mostly white (87%), and a majority college educated (66%). I observe a successful randomization, where individual characteristics are balanced across treatments.

3 Empirical strategy

Following my 2x2 experimental design, here I outline the main specification for the analysis, which has as the baseline group the Homogenous - U.S. National treatment:

$$y_i = \beta_0 + \beta_1 Diverse_i + \beta_2 Foreigner_i + \beta_3 Diverse_i \times Foreigner_i + X_i + \varepsilon_i \tag{1}$$

where y_i correspond to the allocation decision, $Diverse_i$ corresponds to an indicator variable with value 1 if the allocation decision is in the diverse context, $Foreigner_i$ is an

indicator variable with value 1 if the allocation is towards a foreigner, and X_i is a set of collected covariates at the individual level.

I am interested in two main effects captured by this specification. The first corresponds to the effect of a change social context on the allocation towards U.S. nationals, which is captured by β_1 in equation 1. Second, I am interested in the difference across social contexts of the ingroup bias, defined as the differences in the mean allocation towards a U.S. national vs towards a foreigner. The 2x2 experimental design allows for difference-in-difference approach to estimate the effect of a change in social context in the ingroup bias, which is captured by $-\beta_3$ in equation 1. Additionally, β_2 provides the estimated difference between the allocations towards a U.S. national and a foreigner in the homogeneous treatment.

3.1 Hypotheses

The experiment represents a society of $N = 3$, with a DM, a receiver, and a non-receiver. The DM is always from the USA and is exposed to two individuals that can be either a fellow U.S. national, or a foreigner (China or Canada). The analysis follows a simple formal framework, described in Appendix B, and is based on two premises, based on work in economics and psychology.

The first premise is that a decision-maker places a greater weight on the receiver's pay-offs when the receiver is perceived as closer to their own. This draws from research in economics on how individuals behave more altruistic to people that are perceived as part of their ingroup (Chen and Li, 2009), that are socially closer (Leider et al., 2009), and that are perceived as closer (Robson, 2021). The second premise indicates that this perception of social proximity can be affected by the social context, by changing the reference groups. This insight draws from extensive research in psychology suggests that context exert a significant influence on perception (Gold, 2014).[5] Recent research in economics has incorporated these insights into the social domain. For instance, Esponda et al. (2023) shows that inference when assessing new information about an individual, pertaining to a specific group, can be biased by the context, which is represented by

[5]A very influential example corresponds to the Ebbinghaus-Titchener illusion (Titchener, 1901). Two circles of the same size are surrounded by a different context each: the first circle is surrounded by small circles and the second circle is surrounded by big circles. When most observers view these figures, the context affect perceptions of size, through contrasting with the surrounding circles and makes individuals perceive as if they have a different size. This has sparked an extensive literature on the effects of context on perceptions.

the reference group the individual is compared to, in a phenomena closely associated to Kahneman and Tversky's (1972) "representativeness heuristic".

Under the two premises, a change in social context from homogeneous to diverse would affect the reference group of the decision-maker when assessing social proximity. To assess the impact of the reference group, I follow Bordalo et al. (2016), where by contrasting the country of the receiver and the country of the non-receiver, the decision-maker would bias her perceived social proximity towards the receiver in a manner that magnify group differences. This heightened perceived social proximity to the U.S. nationals driven by contrast with the foreigner leads to a rise in ingroup-biased allocations. The model provides the following preregistered hypotheses:

Hypothesis 3.1. *Allocations towards U.S. nationals (foreigners) are higher (lower) in a diverse context with respect to a homogeneous context ($\beta_1 > 0$).*

Hypothesis 3.2. *The ingroup bias in allocations is higher in a diverse context with respect to a homogeneous context ($-\beta_3 > 0$).*

Hypothesis 3.3. *The mechanism behind the effects in hypotheses 3.1 and 3.2, correspond to changes in perceived social proximity as a consequence of changing from a homogeneous to a diverse context.*

4 Results

Figure 9 in Appendix A displays the decision distribution in the allocation task and in the social proximity elicitation for the entire sample. A significant majority, over 80% of participants, designated a positive amount to the receiver. The average allocation stands at approximately 12 USD out of a possible 40 USD, aligning with previous studies on giving (Engel, 2011). In terms of social proximity, more than 40% of the participants indicated the lowest level on the IOS Scale, which is anticipated since the matched participants are unfamiliar to the decision-makers (DMs).

In the subsequent sections, I delve into the impact of transitioning from a homogenous to a diverse social context on the primary outcome: Allocation task. Subsequently, I provide suggestive evidence pointing to perceived social proximity as a key factor influencing the effect of social context on allocations. Leveraging the large-scale sample, I explore variations across population subgroups and find the effect consistently mirrored across demographic groups. Finally, I analyze the potential implications of these main

findings, especially in relation to prior exposure to international diversity over policy preferences.

Figure 3: Mean allocations towards U.S. nationals and foreigners in the homogeneous and diverse contexts.

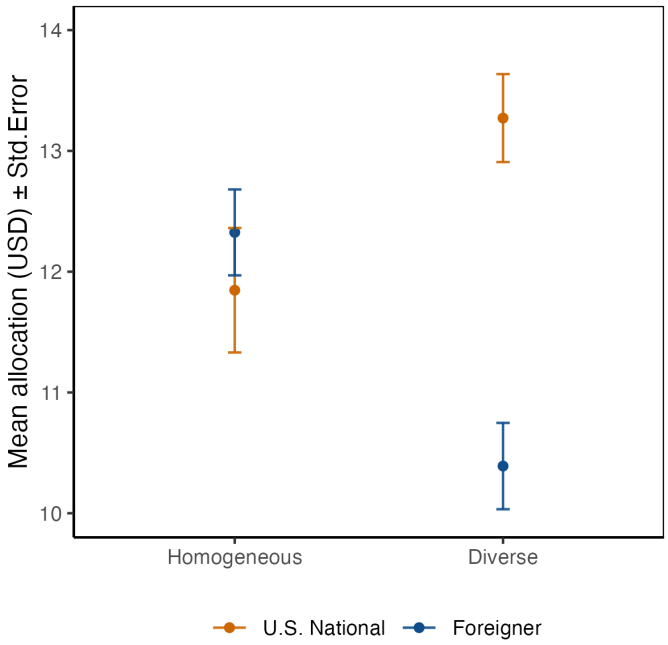


Figure 3 summarizes the allocations of the decision-makers. The patterns underline the influence of social context on prosocial behavior. In a diverse setting, the mean allocation to a U.S. national surpasses that in a homogeneous context. On the other hand, allocations to foreigners decrease in diverse settings. Thus, the ingroup bias in allocations in a diverse setting is broader than in a homogeneous context. Notably, the bias is negligible in a homogeneous setting. Collectively, these findings suggest that diversity skews allocations in favor of U.S. nationals, and that social context is key for the emergence of ingroup bias.

Table 1 offers regression estimates following specification 1, with allocations in USD as the dependent variable, to represent the treatment effects of exposure to diversity in our 2x2 design. The variable "Diverse" indicates the diversity treatment's effect on allocations to U.S. nationals. The average allocation to an American in a diverse setting exceeds its counterpart in a homogeneous setting by 1.4 USD (p=0.03), representing an approximate 12% increase. This increase remains robust even when controlling with out set of collected individual-level characteristics. A country-wise breakdown of the

Table 1: OLS estimates of the regression on the allocation.

	Allocation in USD					
	Pooled sample		Foreigner: Canada		Foreigner: China	
	(1)	(2)	(3)	(4)	(5)	(6)
Diverse	1.425**	1.415**	1.631**	1.537**	1.223*	1.214*
	(0.613)	(0.600)	(0.712)	(0.700)	(0.726)	(0.715)
Foreigner	0.479	0.557	1.059	1.057	−0.122	0.027
	(0.608)	(0.593)	(0.698)	(0.684)	(0.723)	(0.706)
Diverse x Foreigner	−3.360***	−3.343***	−2.675***	−2.551***	−3.992***	−4.008***
	(0.794)	(0.778)	(0.994)	(0.973)	(1.027)	(1.011)
Control mean	11.85	4.75	11.85	3.88	11.85	5.64
Controls	No	Yes	No	Yes	No	Yes
Observations	2,808	2,808	1,601	1,601	1,604	1,604
R^2	0.012	0.053	0.005	0.054	0.021	0.066

Note: *p<0.1; **p<0.05; ***p<0.01

This table presents the results from an OLS regression with robust S.E. The variable Diverse takes value 1 when the participant is in the diverse context and 0 otherwise, and the variable Foreigner takes value 1 when the allocation is towards a foreigner or 0 otherwise. Diverse x Foreigner shows the estimate of the interaction. Columns 1 and 2 shows the estimates by using the entire sample. Column 3 and 4 estimate the coefficients using the subsample where the foreign country was specifically Canada, and columns 5 and 6 when the foreign country was China. The even columns show specifications that control for all covariates measured, except for the elicited social proximity.

two foreign countries (China and Canada) confirms the consistency of this result.

The allocation decisions are visually represented in Figures 11 of Appendix A. It seems the shift in mean allocation towards U.S. nationals is driven more by distribution changes than by the number of participants allocating positively. A chi-squared test shows that there is no difference in the proportion of respondents that give a positive amount across contexts.

The 2x2 experimental design facilitates a difference-in-differences approach that allow us to study the effect of exposure to diversity in the ingroup bias. The effect of diversity on the ingroup bias is given by the opposite of the interaction effect of Table 1. The regressions estimates show a significant widening of the ingroup bias in diverse contexts. This divergence is consistent, regardless of the foreign country being evaluated ($p<0.01$ for both China and Canada). This results in an ingroup bias in a diverse society where allocations towards fellow U.S. nationals are 28% higher than towards foreigners. This confirms that diversity not only bolsters allocation towards U.S. nationals

but also accentuates disparities in behaviors towards Americans and foreigners, which is intensified by a decrease in the allocations towards foreigners.

Additionally, I aimed to discern any allocation disparities towards U.S. nationals and foreigners in both contexts. The coefficient "Foreigners" reflects this differential in the homogeneous context alone. Notably, there is no discernible gap in the homogeneous context (p=0.54). This holds true even when examining the differential specifically for Canada or China. A joint-F test for null hypothesis of both countries as well as individual null hypothesis tests for each country show no positive ingroup bias in a homogeneous context (p>0.3). Furthermore, I analyze the bias for each partition in the set of covariates. Figure 12 in Appendix A shows that across subsamples of the collected characteristics, there is no positive ingroup bias.

Result 1. *Exposure to diversity increases ingroup bias, where individuals allocate more to U.S. nationals and less to foreigners relative to a homogeneous context. Notably, decision-makers in a homogeneous contexts do not exhibit such bias.*

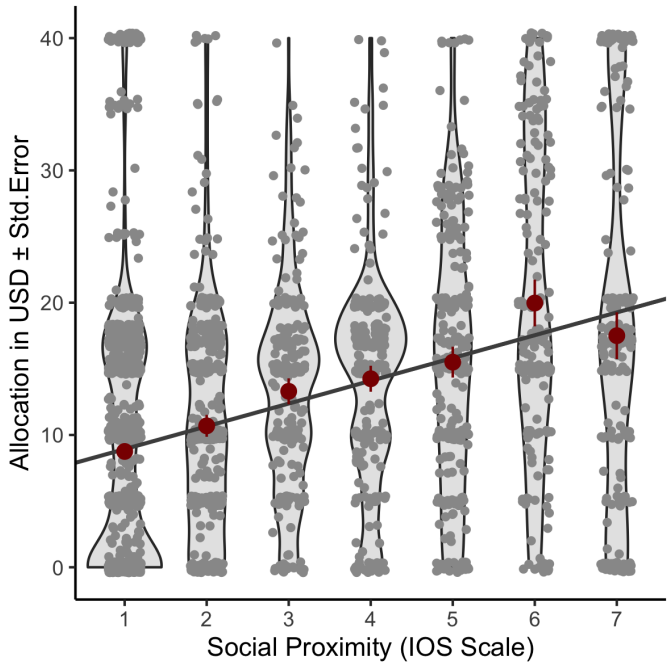
These findings highlight that social context is key for the emergence and strength of ingroup bias. The results do not contradict previous work on identity effects, as in most studies that investigate ingroup bias, to the best of my knowledge, the experimental design is either within-subject, or in a setting that highlights the diversity with competing identities (Charness and Chen, 2020; Shayo, 2020). However, the fact that informing individuals about group membership alone does not generate identity effects highlights the importance on social context in the origins of ingroup favoritism.

Appendix A introduces various robustness checks of the analysis of treatment effects for on the allocation task. To address the concern of whether the effects on the mean allocation are driven by extreme outcomes, I analyze the outcomes using non-parametric Mann-Whitney tests and Randomization Inference tests (Young, 2019), where I find that the results are robust (p<0.04) for a two-sided test in the pooled sample (see 11 for the histogram of allocations by treatment) over the effects over the allocations towards both, U.S. nationals and foreigners. Moreover, we run the same specification 1 on a subsample of the population after excluding the individuals that allocated all their pot (40 USD) and the results persist (see Table 5).

4.1 Mechanisms

As posited in section 3, the first premise of our formal framework theorized that allocation decisions relate positively with perceived social proximity. Figure 4 shows the distribution and mean allocation of a respondent towards an individual with a fixed level i in the IOS scale, for all levels of the IOS scale $i = 1, ..., 7$, where 1 is not close and 7 is very close. The figure establishes a clear correlation: the closer respondents felt to a participant, the more they allocated. The statistical relationship is compelling ($p<0.01$), with tests affirming that as perceived proximity increases, allocations follow suit. Moreover, non-parametric Mann-Whitney tests show for each consecutive comparison of levels of social proximity, that the distributions are shifted towards higher allocations, the closer the selected participant is perceived ($p<0.05$).

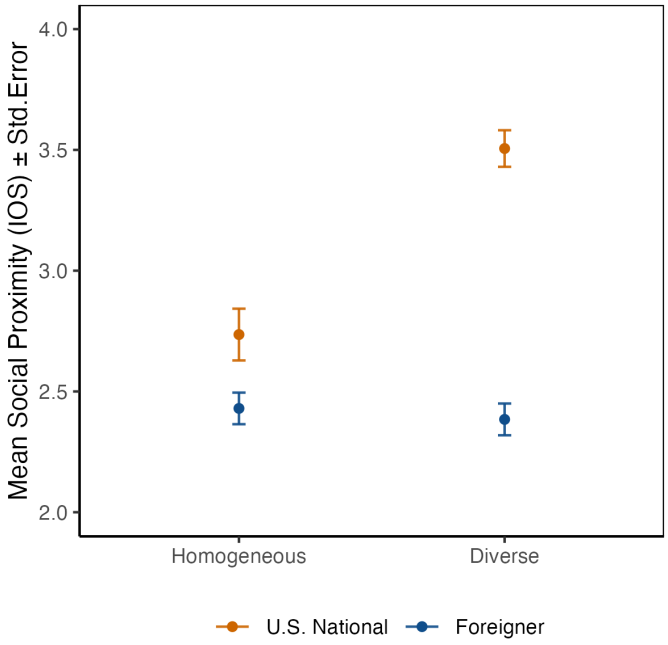
Figure 4: Distribution and mean allocations for each level of social proximity using the IOS Scale (from 1-7).



The second premise suggests that changes in the social context affects perceived social proximity, in a way consistent with a contrast between the country of the receiver and the non-receiver. Figure 5 suggests that perceived social proximity exhibits a similar pattern as allocation decisions, which were shown in Figure 3. Columns 1 and 2 of Table 2 replace the dependent variable with the social proximity measure in specification 1.

Coefficient "Diverse" reveals an increase in perceived proximity towards U.S. nationals in diverse contexts. Moreover, the opposite of the interaction effect between "Diverse" and "Foreigner" demonstrates that showcasing diverse identities increases the ingroup bias in social proximity. Therefore, the treatment triggers the same effect for the U.S. nationals in both outcomes, the allocation and the social proximity. However, the social proximity to the foreigners is not influenced by exposure to diversity.

Figure 5: Mean level of social proximity by treatment group using the IOS Scale (from 1 to 7).



I now turn my attention to the role of perceived social proximity as a mediator of the treatment effect in the allocation decision. Columns 3 to 6 in Table 2 display the regression estimates based on specification 1, with the allocation serving as the dependent variable. Interestingly, when perceived social proximity is included as a control variable, the effect of diversity on the allocations towards U.S. nationals vanishes. Additionally, the coefficient capturing the interaction effect diminishes considerably. While this coefficient becomes significantly smaller, it retains its significance. The explanatory power of the linear regression also sees a notable boost, which aligns with strong positive relationship between perceived social proximity and allocation.

Result 2. *Evidence suggests that the increase in ingroup bias in allocations is driven by changes in perceived social proximity caused by exposure to diversity.*

Table 2: OLS estimates of the regression on the social proximity and analysis of mechanism of effects on allocation.

	Social Proximity (IOS)		Allocation in USD			
	(1)	(2)	(3)	(4)	(5)	(6)
Diverse	0.770***	0.737***	1.425**	1.415**	0.102	0.195
	(0.126)	(0.125)	(0.613)	(0.600)	(0.587)	(0.577)
Foreigner	−0.306***	−0.322***	0.479	0.557	1.004*	1.090*
	(0.119)	(0.117)	(0.608)	(0.593)	(0.572)	(0.562)
Diverse x Foreigner	−0.816***	−0.775***	−3.360***	−3.343***	−1.958***	−2.061***
	(0.157)	(0.155)	(0.794)	(0.778)	(0.753)	(0.740)
Proximity					1.719***	1.654***
					(0.107)	(0.106)
Control mean	2.74	2.46	11.85	4.75	7.14	0.67
Controls	No	Yes	No	Yes	No	Yes
Observations	2,808	2,808	2,808	2,808	2,808	2,808
R^2	0.055	0.088	0.012	0.053	0.122	0.151

Note: *p<0.1; **p<0.05; ***p<0.01

This table presents the results from an OLS regression with robust S.E., using the pooled sample, on the social proximity, which takes value 1 to 7. The variable Diverse takes value 1 when the participant is in the diverse context and 0 otherwise, and the variable Foreigner takes value 1 when the allocation is towards a foreigner or 0 otherwise. Diverse x Foreigner shows the estimate of the interaction, what I call the diff-in-diff estimate. Columns 1 and 2 shows the coeficients of the regression with dependent variable Social Proximity. Column 3 to 6 estimate the coefficients with dependent variable Allocation and studies the role of social proximity in explaining the effects on the allocation.

The influence of social proximity as a mediator of the effect of social context on prosocial behavior stands strong under different types of analysis. Table **??** presents mediation analysis following Imai et al. (2010) and Heckman and Pinto (2015). This analysis leverages exogenous variation in two dimensions: the group membership of the receiver and the overarching social context. The aim is to determine if social proximity can account for a portion of the treatment effect on allocation. Both the indirect and direct effects of the treatment remain significantly positive after introducing social proximity as a mediator.

4.1.1 Discussion and alternative mechanisms

While this paper presents compelling evidence for social proximity influencing ingroup allocations, it is vital to consider other possible drivers. In our experimental framework, every dollar allocated to the receiver creates a disparity between the receiver and the non-receiver. It's plausible that individuals display heightened aversion to inequality when faced with two members of the same country, compared to when they're distinct. Although Figure 5 and table 2 reveal a trend in line with this hypothesis for U.S. nationals, this isn't the case for the foreigners allocation, where a DM appears more tolerant of inequality in homogeneous contexts than in diverse ones. Moreover, if this alternate mechanism were solely responsible for the effects on U.S. nationals, I would expect a significant residual of the treatment effect in the "Diverse" coefficient of 2, after controlling for percceived social proximity, which does not hold.

As depicted in Figure 5, social context has no influence on the social proximity towards foreigners members. Figure 11b highlights that most participants indicated the lowest level of perceived social proximity, 1, towards foreigners in the homogeneous context. This suggests that a potential explanation of the null effect of context on social proximity might be associated to reaching the bound in the choice set. Regardless of the explanation for the negligible effect, changes in social proximity cannot account for the observed effects on allocations towards foreigners. This leaves a segment of the ingroup bias in allocations unexplained.

There are two alternative explanations. First, in the allocation decision, the weight that the DM puts on the receiver's pay-offs might be dependent on the relative proximity, defined as the difference between proximity between the U.S. nationals and foreigners, instead of the absolute proximity to the receiver. This relative proximity seems to surge in diverse settings, as evident in Figure 5. This intuition relates to the work

of social psychology on social categorization theory, where individuals when classifying others as ingroup or outgroups, they use a *meta-contrast principle* which maximizes difference across groups and minimize distance within groups (Tajfel et al., 1979; Turner et al., 1987). An adaptation of the conceptual framework presented in Appendix B, where absolute proximity to the receiver is substituted for relative proximity, provides us with predictions consistent with the empirical findings of the experiment. However, individual-level data on relative proximity is missing, limiting our ability to discern any patterns.

A second theory proposes variable levels of inequality aversion based on the social context and which group benefits from the inequality. My research indicates that allocations shift in favor of the ingroup, creating disparities. For allocations towards U.S. nationals, changes in social proximity leave no residual of the effect to be solely attributed to inequality aversion. Testing this for allocations towards foreigners is not feasible.

4.2 Heterogeneity analysis

By leveraging the representative sample in this study, I delve into the heterogeneity of effects across various population subgroups. Figure 6 shows the coefficients of the effects of diversity on (i) the allocations towards U.S. nationals, (ii) the ingroup bias in allocations, and (iii) the perceived social proximity towards U.S. nationals, which serves as a mechanism in the effects, across subgroups formed from the collected set of demographic characteristics. While there are no differences accross groups in the coefficients presented, they are systematically consistent with the hypotheses, and present across any subgroup of our set of covariates. This correspond to suggestive evidence that the results seem to reflect a general feature in decision-making.[6]

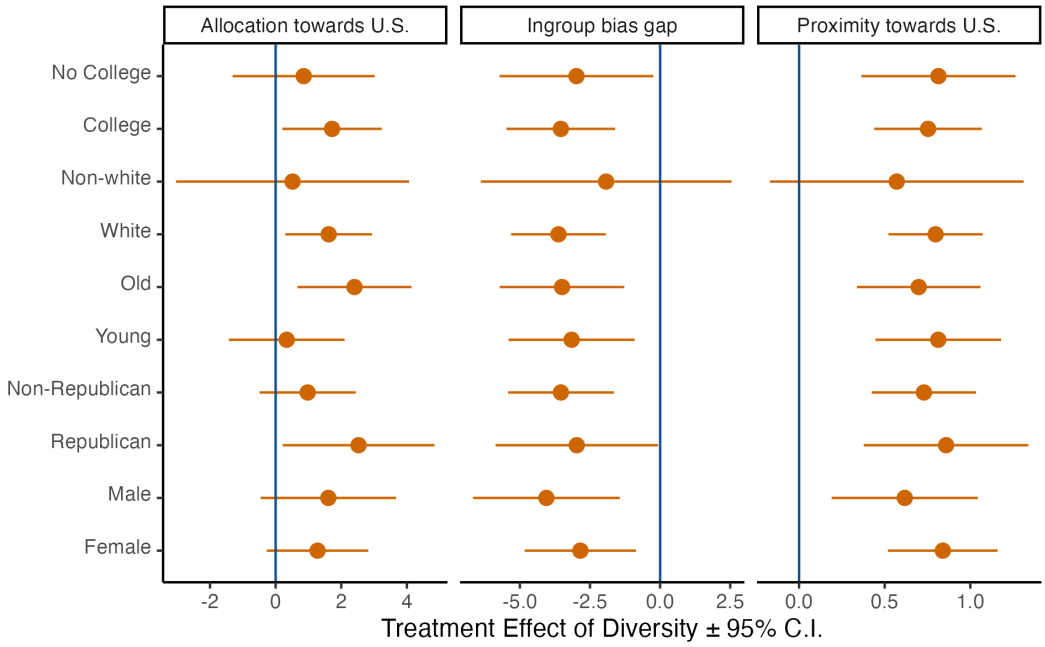
Result 3. *The effect of diversity on allocations and perceived social proximity are consistent and robust, irrespective of population subgroups.*

4.3 Implications

I conduct an exploratory analysis to assess policy relevant implications of the experimental findings, focusing on preferences over redistributive policies. For this analysis,

[6]Note that the analysis by subgroup splits the sample, which eventually reduces power to detect statistical impacts, however, in all three variables I find at least one subgroup to have a significant positive effect ($p<0.05$) with correction for multiple hypothesis testing.

Figure 6: Coefficients of the effect of diversity on the three main outcomes of analysis: allocations towards U.S. nationals, ingroup bias and proximity to U.S. nationals, by subgroup of the population.



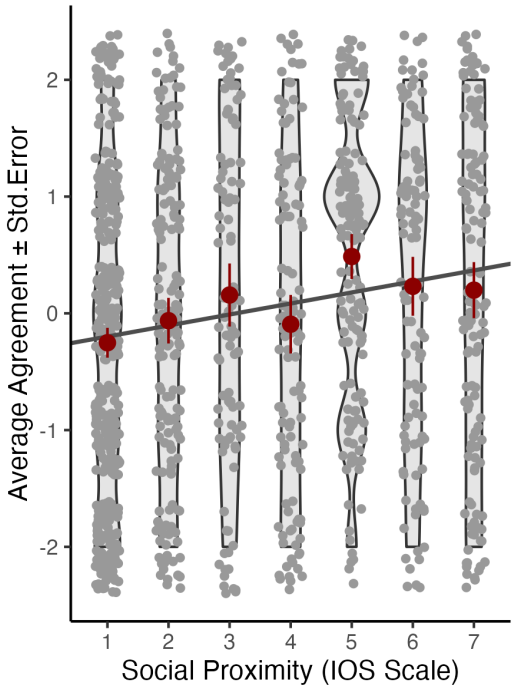
I am interested in understanding whether the effects in social context over behavior and perceptions towards U.S. nationals carry over policy outcomes targeted to Americans.

Policy views. I elicited preferences over a redistributive policy, where participants were asked to indicate their agreement with a statement prioritizing nationwide over local redistribution of a tax that was collected locally, in the form of welfare payments. The agreement scale is a 5-point scale with a value of -2 for "Strongly Disagree" and a value of 2 for "Strongly Agree".

Around 36% of participants agree with prioritizing nationwide over local redistribution on the baseline. Aimed to relate our behavioral measures with agreement on this policy, Figure 7 shows how agreement with prioritizing nationwide redistribution relates to perceived social proximity towards a U.S. national, as presented in the experiment. The closer a U.S. decision-maker feels towards an anonymous fellow U.S. national, the more she supports prioritizing nationwide redistribution.

Now, I am interested in evaluating whether the increased perceived social proximity towards fellow U.S. nationals generated by exposure to diversity carries over into the political domain. As a reminder, exposure to a diverse context corresponds to a

Figure 7: Distribution and mean level of agreement to prioritizing nationwide over local redistribution, for each level of social proximity using the IOS Scale (from 1-7).



decision-maker facing a fellow U.S. national and a participant living in a different country (China or Canada), who does not benefit from the hypothetical policy. Table 3 shows the effect of prior exposure to diversity over the average agreement to the statement that prioritizes nationwide over local redistribution. The table reveals that a diverse context bolsters support for nationwide policies relative to a homogeneous context. The effect size corresponds to an increase in the average agreement of 0.11, on a 5-point scale. These results are robust to alternative specification choices, such as employing an ordered probit or assessing the effect on the percentage that agrees, as shown in Table 8 in Appendix A. Furthermore, Figure 13 in the Appendix A also proves that there is no heterogeneity in the effects by subgroups of the population, and the effect is consistent across subgroups.

To associate the results on policy views with the findings of the main experiment, I explored the role of perceived social proximity in explaining policy support. Column 3 of Table 8 shows that when controlling for social proximity, the effect of diversity on support for nationwide welfare payments vanishes. This finding suggests that for the individuals who had an increase in perceived social proximity towards U.S. nationals after

Table 3: OLS estimates of the regression on the level of agreement of prioritizing nationwide over local redistribution of a tax.

	Support for Nationwide Welfare payments		
	(1)	(2)	(3)
Diverse	0.101*	0.112**	0.070
	(0.052)	(0.050)	(0.049)
Male		0.194***	0.146***
		(0.053)	(0.053)
Republican		−0.659***	−0.657***
		(0.055)	(0.054)
Older than 50		−0.324***	−0.266***
		(0.050)	(0.050)
College		−0.017	−0.023
		(0.051)	(0.051)
Proximity			0.105***
			(0.012)
Controls	No	Yes	Yes
Observations	2,808	2,808	2,808
R^2	0.001	0.085	0.109
Note:			*p<0.1; **p<0.05; ***p<0.01

This table presents the results from an OLS regression with robust S.E., where the dependent variable corresponds the level of agreement over the redistributive policy, which is a 5-point scale ranging from -2 to 2. The variable Diverse takes value 1 when the participant is in the diverse context and 0 otherwise.

exposure to context, the effects carry over into support for nationwide redistribution.

Collectively, these results provide evidence of the treatment effects of exposure to diversity on perceived social proximity carrying over into policy views, in a manner consistent with heightened ingroup favoritism towards USA nationals. An increased social proximity generated by exposure to diversity translates into higher support for national policies. These findings align with empirical work in political science and economics on the effects of migration or war in shaping ingroup boundaries and nation-building, where the emergence of an opposing third party amplifies nationalistic sentiments (Fouka and Tabellini, 2022).

Result 4. *The increase in perceived social proximity towards fellow U.S. national caused by a change in social context carries into the political domain, making participants support more national redistribution.*

5 Discussion and next steps

In a world increasingly marked by diversity in neighborhoods, schools, and workplaces, understanding the effects of changes in a social context on behavior becomes crucial. This paper explored how changes in social context influenced perceptions of social proximity and consequent prosocial behavior. Using a novel experimental approach, I highlight role of social context on the emergence of ingroup bias with implications over policy preferences.

Four main findings emerge. First, the exposure to a diverse context intensifies ingroup bias in allocations favoring U.S. nationals. Second, shifts in perceived social proximity— caused by changes in the social context— correspond to a mechanism in the effects on allocations. Third, the effects of social context on both allocations and perceived social proximity are robust and consistent across subgroups of the population. Finally, the increase in perceived social proximity towards U.S. nationals triggered by exposure to diversity carries over political attitudes, increasing support for nationwide redistributive policies. Altogether, these findings offer crucial insights into how social contexts can affect behavior in political and economic decisions.

The fact that mere exposure to diversity can increase ingroup bias in prosocial behavior, highlights that if a social planner's goal is to increase social cohesion or intergroup cooperations, it is important to avoid situations in which social contexts correspond to mere exposure to diversity, as my findings suggest biases might emerge. The social

planner should generate conditions such that the proposed perceptual mechanism has a diminished role in exacerbating differences in behavior towards ingroup or outgroup members.

However, more exploration regarding different ways in which context can affect prosocial behavior is yet to be explored. This paper abstracts from social signaling concerns in the allocation decisions. However, whether the receiver or non-receiver is aware or not about the allocation decision made by the decision-maker, might affect their behavior, and affect the intensity of the results, which corresponds to an important extension to explore.

In this paper, I isolate the effect of mere exposure to diversity on interactions, and provide a mechanism and measure through which prosocial behavior is affected by social context. The experimental design focuses on immediate and short term exposure. Future work should address how the mechanism proposed evolves when the exposure is long term. Moreover it is also key to understand, how perceived social proximity towards both: ingroups and outgroups, changes when contact in the society happens.

Further explorations that attempt to generalize the effects can be performed. For instance, studying whether the effects differ or persist across other types of identities: e.g., ethnicity, religion, etc. It is possible to make use of lab-in-the-field tools, in schools or universities to understand whether the results are robust to settings where identity is not explicitly indicated. Furthermore, another direction is to extend the scope of study, and evaluate the effects of social context on other types of prosocial behavior, such as trust and cooperation. Implications of a more complete set of choices can speak to a wider set of real life settings, and give a better understanding over the relevance of social context on prosocial behavior.

Finally, this study provides a building stone for future research on the effects on context on decision making. The experimental design allows the researcher to cleanly study context effects in a wide array of decision sets, particularly within the framework of contrast biased evaluations.

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A Additional tables and figures

Figure 8: Mean level of perceived closeness towards a person from the USA, from China and from Canada in the no-contrast context.

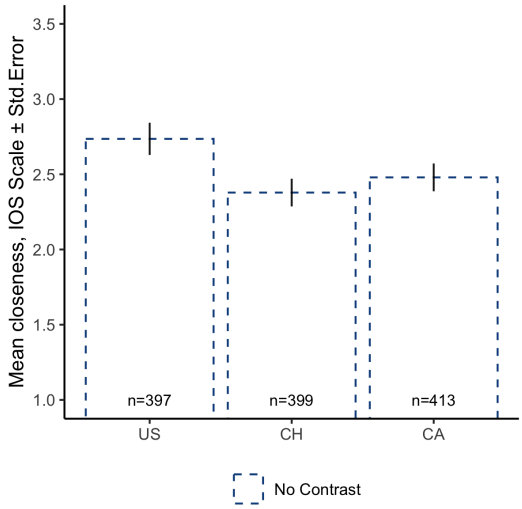
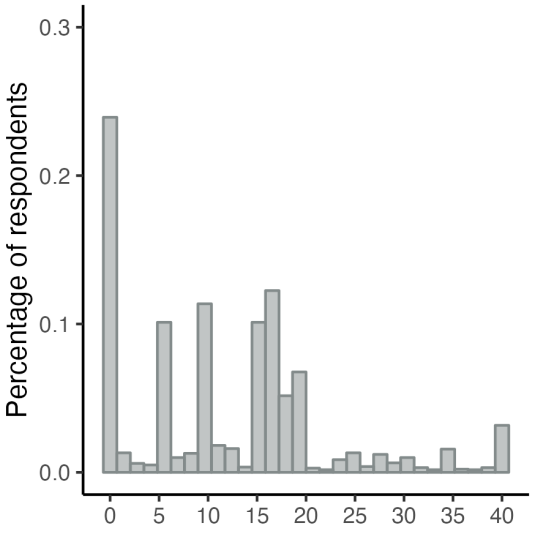


Figure 9: Histograms of the allocation task (left) and the social proximity elicitation (right).



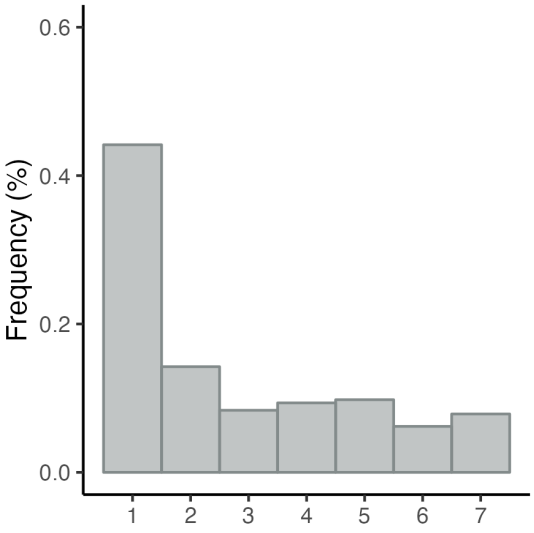
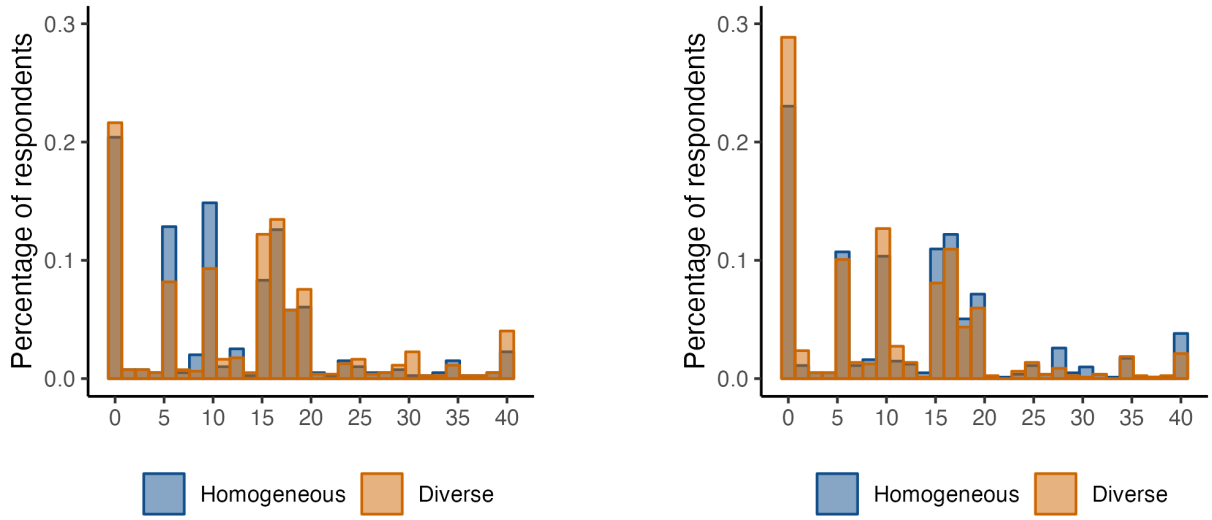


Table 4: Descriptive statistics: number of observations and means across collected outcomes.

		Outgroup: China			Outgroup: Canada		
	IN-IN	IN-OUT	OUT-IN	OUT-OUT	IN-OUT	OUT-IN	OUT-OUT
Age	51.65	49.84	50.77	51.05	49.86	50.98	51.34
	(15.08)	(15.46)	(15.16)	(14.54)	(15.23)	(15.07)	(14.94)
Female	0.65	0.62	0.65	0.60	0.63	0.65	0.61
	(0.48)	(0.49)	(0.48)	(0.49)	(0.48)	(0.48)	(0.49)
College	0.68	0.64	0.64	0.66	0.66	0.65	0.69
	(0.47)	(0.48)	(0.48)	(0.48)	(0.47)	(0.48)	(0.46)
White	0.89	0.86	0.86	0.88	0.86	0.87	0.86
	(0.32)	(0.34)	(0.34)	(0.33)	(0.35)	(0.34)	(0.34)
Repub.	0.28	0.28	0.34	0.30	0.32	0.35	0.29
	(0.45)	(0.45)	(0.47)	(0.46)	(0.47)	(0.48)	(0.45)
Univers.	-0.23	0.11	-0.00	-0.29	-0.21	-0.13	-0.46
	(2.90)	(2.85)	(3.17)	(2.87)	(2.94)	(3.04)	(3.00)
Altruism	7.81	7.75	7.75	7.54	7.87	7.77	7.79
	(2.50)	(2.32)	(2.48)	(2.56)	(2.35)	(2.52)	(2.37)
Allocation	11.85	13.07	8.96	11.72	13.48	11.86	12.91
	(9.66)	(10.79)	(9.89)	(10.70)	(10.33)	(9.52)	(10.17)
N	397	401	407	399	394	397	413

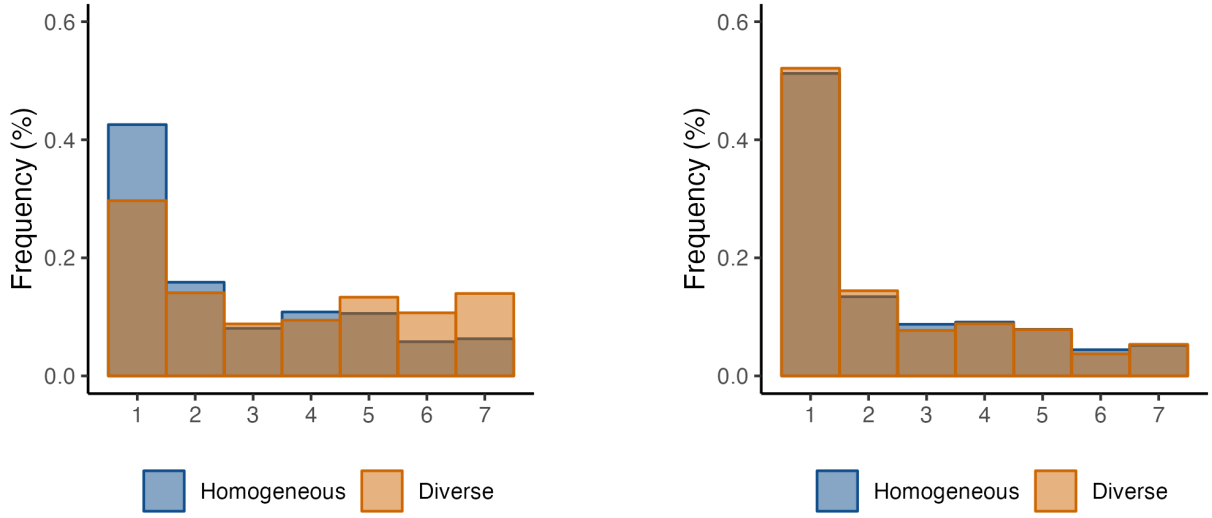
Notes: The table shows the mean value and standard deviation of each collected variable for each treatment group. Female, College, White and Republican are indicator variables. Age is measured in years. Universalism is measured in a scale from -2 to 2, where -2 represents perfect non-universalist and 2 perfect universalist. Altruism is a scale from 0 to 10, were 0 is low willigness to give, and 10 is high willingness to give. Finally, Allocation is the amount of USD given to the selected participant, with the choices ranging from 0 to 40 USD.

Figure 10: Histogram of the allocations towards U.S. nationals and foreigners in each social contexts: homogeneous and diverse.



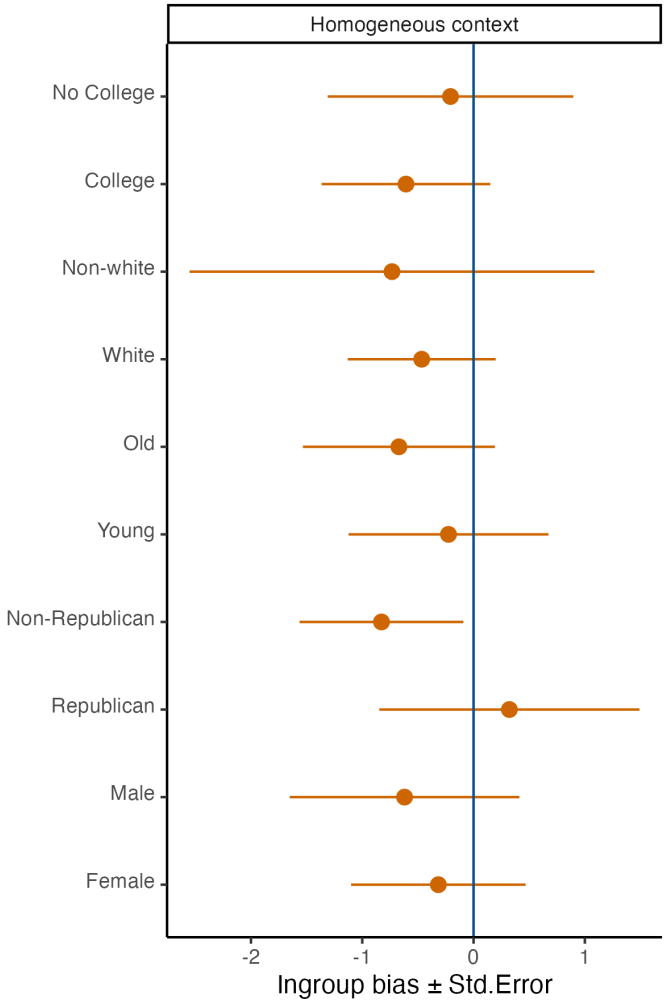
(a) U.S. nationals **(b)** Foreigners

Figure 11: Histogram of the allocations towards U.S. nationals and foreigners in each social contexts: homogeneous and diverse.



(a) U.S. nationals **(b)** Foreigners

Figure 12: Ingroup-outgroup gap in the non-contrasting context, for the pooled sample, for each subsample of covariates collected.

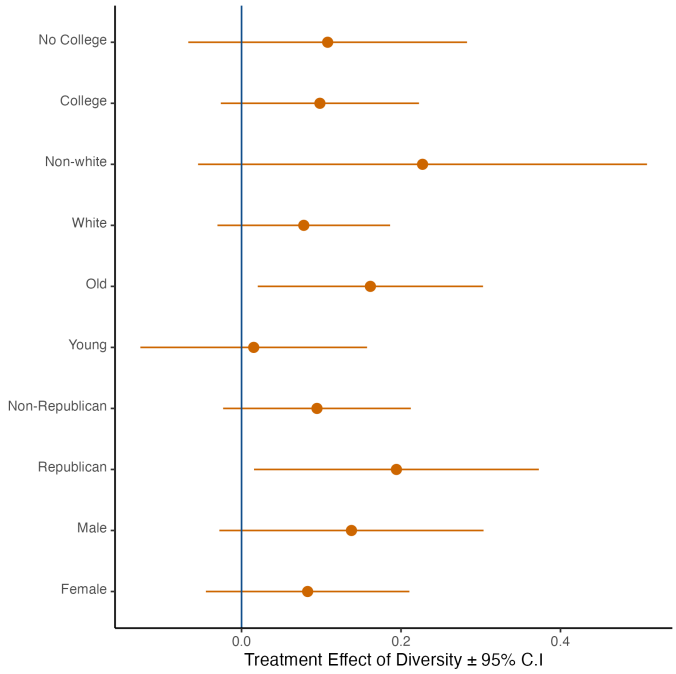


Notes: In this figure we show the point estimates of an OLS regression of the allocation on a dummy that takes value one if the allocation was for the ingroup and 0 if it was for the outgroup, for the non-contrast sample, for each subsample of covariates collected. We show the point estimates ± Robust S.E., for the regression without controls.

Table 5: OLS regression under different subsamples and specifications.

	Positive giving		Allocation in USD	
	(1)	(2)	(3)	(4)
Diverse	−0.012	−0.011	0.957*	0.971*
	(0.025)	(0.024)	(0.556)	(0.542)
Foreigner	−0.026	−0.021	0.033	0.139
	(0.025)	(0.025)	(0.550)	(0.536)
Diverse x Foreigner	−0.046	−0.049	−2.433***	−2.476***
	(0.033)	(0.033)	(0.718)	(0.702)
Control mean	0.8	0.55	11.19	5.07
Controls	No	Yes	No	Yes
Observations	2,808	2,808	2,719	2,719
R^2	0.006	0.036	0.010	0.048
Note:				*p<0.1; **p<0.05; ***p<0.01

Figure 13: Effect of exposure to diversity on support for national welfare payments, for each subsample of covariates collected.



Notes: In this figure we show the point estimates of an OLS regression of the agreement level on a dummy that takes value one if the participant was exposed to diversity, for each subsample of covariates collected. We show the point estimates ± Robust S.E., for the regression without controls.

Table 6: The additional treatment effect of diversity on the allocation by respondents who first indicated Social Proximity.

	Allocation in USD		
	Pooled sample	Foreigner: Canada	Foreigner: China
	(1)	(2)	(3)
Diverse	1.150	0.898	1.360
	(0.853)	(0.989)	(0.994)
Second	−0.055	−0.055	−0.055
	(0.973)	(0.973)	(0.973)
Foreigner	0.687	1.224	0.094
	(0.857)	(0.976)	(1.031)
Diverse x Second	0.560	1.383	−0.313
	(1.229)	(1.422)	(1.460)
Foreigner x Second	−0.416	−0.342	−0.416
	(1.219)	(1.399)	(1.450)
Diverse x Foreigner	−2.851**	−1.303	−4.532***
	(1.113)	(1.392)	(1.431)
Diverse x Foreigner x Second	−1.043	−2.812	1.085
	(1.590)	(1.982)	(2.062)
Control mean	11.88	11.88	11.88
Controls	No	Yes	No
Observations	2,808	1,601	1,604
R^2	0.013	0.008	0.022

Note: *p<0.1; **p<0.05; ***p<0.01

Table 7: The additional treatment effect of diversity on the social proximity by respondents who first indicated the allocation decision.

	Social proximity (IOS)		
	Pooled sample	Foreigner: Canada	Foreigner: China
	(1)	(2)	(3)
Diverse	1.190***	0.872***	1.559***
	(0.173)	(0.199)	(0.207)
Second	0.139	0.139	0.139
	(0.198)	(0.198)	(0.198)
Foreigner	−0.239	−0.305*	−0.176
	(0.163)	(0.182)	(0.193)
Diverse x Second	−0.830***	−0.555*	−1.163***
	(0.252)	(0.296)	(0.295)
Foreigner x Second	−0.137	0.087	−0.377
	(0.238)	(0.271)	(0.271)
Diverse x Foreigner	−1.150***	−0.573**	−1.745***
	(0.217)	(0.271)	(0.281)
Diverse x Foreigner x Second	0.660**	0.134	1.208***
	(0.313)	(0.397)	(0.393)
Control mean	2.67	2.67	2.67
Controls	No	Yes	No
Observations	2,808	1,601	1,604
R^2	0.064	0.033	0.092

Note: *p<0.1; **p<0.05; ***p<0.01

Table 8: Effects of exposure to diversity on policy views under different specifications, Probit and OLS regression with Robust Standard errors.

	Prioritize Nationwide Welfare payments					
	Average agreement			Percentage that agrees		
		ordered probit			*OLS*	
	(1)	(2)	(3)	(4)	(5)	(6)
Diverse	0.075*	0.087**	0.054	0.039**	0.043**	0.029
	(0.040)	(0.040)	(0.041)	(0.019)	(0.018)	(0.018)
Male		0.150***	0.113***		0.104***	0.089***
		(0.042)	(0.042)		(0.019)	(0.019)
Republican		−0.541***	−0.547***		−0.169***	−0.168***
		(0.044)	(0.045)		(0.020)	(0.019)
Older than 50		−0.257***	−0.214***		−0.085***	−0.066***
		(0.041)	(0.041)		(0.018)	(0.018)
College		−0.014	−0.018		0.031	0.029
		(0.042)	(0.043)		(0.019)	(0.019)
Proximity			0.086***			0.034***
			(0.010)			(0.004)
Control mean	0.08	0.09	0.05	0.36	0.23	0.16
Controls	No	Yes	Yes	No	Yes	Yes
Observations	2,808	2,808	2,808	2,808	2,808	2,808
R^2				0.002	0.063	0.083

Note: *p<0.1; **p<0.05; ***p<0.01

B Conceptual framework

Consider a DM from the U.S. facing a society composed of herself (i) and two other individuals (j and $-j$). A DM must allocate resources with only one randomly selected individual: receiver j. In this society, each individual $k \in \{i, j, -j\}$ has group membership $g_k \in \{N, F\}$, which correspond to U.S. national (N) and foreigner (F). I can define G_s to be the set of existing groups that the DM is exposed to in a society s. I consider two types of social contexts characterized by G_s: a homogeneous context, where either $G_s = \{N, N\}$ or $G_s = \{F, F\}$, and a diverse context, where $G_s = \{N, F\}$.

The conceptual framework is based on two premises established in the literature. The first premise corresponds to the fact a decision-maker places a greater weight on the receiver's pay-offs when the receiver is perceived as closer to their own, which draws from research in economics on how individuals behave more altruistic to people that are perceived as part of their ingroup (Chen and Li, 2009), that are socially closer (Leider et al., 2009), and that are perceived as closer (Robson, 2021). The second premise indicates that this perception of social proximity can be affected by the social context, by changing the reference groups. This insight draws from extensive research in psychology suggests that context exert a significant influence on perception (Gold, 2014). In what follows, I present a formalization of each premise, and subsequently, I will present a series of hypothesis building from the premises.

B.1 Premise 1. Allocations and perceived social proximity are positively correlated

In the experiment, the selected players will earn together a total of \$50 dollars, which are initially distributed as \$40 for DM i, \$5 for receiver j and \$5 for non-receiver $-j$. The DM (i) chooses the amount that she wants to redistribute from her initial endowment towards the selected receiver (j). To inform the analysis, we assume that the DM is maximizing a utility function that depends on her own pay-off and what the selected individual j receives. The pay-off that the other individual $k \neq i, j$ in the society receives is not relevant for the DM, as the concerns over the earnings of any k are assumed to be separable, and the amount k receives is fixed. Following Cappelen et al. (2007) the optimization problem is represented as follows:

$$\max_{x_j} U_i(x_i, x_j) = x_i - \frac{w(\hat{\gamma}_j)}{2}(x_j - F_j)^2 \qquad \text{s.t.} \quad x_i + x_j = 45, \quad x_j \geq 5$$

where x_i and x_j correspond to the allocations that i and j receive, respectively; $w(.)$ represents the weight player i puts on allocating the fair outcome towards player j, relative to her selfishness; and $\hat{\gamma}_j$ is the perceived social proximity towards participant j, which we define as the probability of j being a part of the ingroup. Finally, F_j represents what the DM considers the fair income to j, which is assumed to be independent of the treatment manipulation. Therefore, if the solution to the problem above is interior, the optimal allocation will be given by the following equation:

$$x_j^* = F_j - \frac{1}{w(\hat{\gamma}_j)} \tag{2}$$

which indicates that the optimal allocation depends on what i consider fair to give to j and the weight i gives on the fair allocation. Premise 1 is represented under the following assumption:

P1. $\frac{\partial x^*}{\partial \hat{\gamma}_j} > 0$.

This assumption indicates that the weight player i puts on allocating the fair outcome towards j depends positively on how close i feels towards j. This imply that an increase in perceived social proximity $\hat{\gamma}_j$ generates an increase in allocation x_j^*.

B.2 Premise 2. Social context affects perceived social proximity

From premise 1, the DM behaves more prosocial to j, if the DM considers j to be an ingroup. However, the DM does not know with certainty whether j is an ingroup (in) or not. Instead, the DM takes into account in her decision the probability of individual j to be an ingroup, denoted by $\gamma_j = f(in|g_j)$, which we define as the social proximity j, and we assume that $f(in|N) > f(in|F)$, as the DM is from the U.S.[7] In my set-up, for a DM to determine how much to give to others, she must assess the perceived social proximity of j within a society where she is exposed to other individuals, where each individual of the society has group membership $g_k \in G_s$.

[7]The decision of interpreting social proximity as a probability comes from simplicity in adapting Esponda et al. (2023) framework into this setting. The rational resonates with the assumption that individuals care more about people that they consider an ingroup, relative to unknown others (Chen and Li, 2009).

I consider a DM that might suffer from contrast-driven biases in perceptions in their assessment of perceived social proximity of j, which will depend on the group distribution of the society G_s. The framework draws from Kahneman and Tversky's (1972) "representativeness heuristic", where a decision-maker can form distorted beliefs about a target group by overweighting its representative types. I incorporate this formally following Bordalo et al. (2016) representativeness measure given by the likelihood ratio: $R(in, g_j, g_{-j}) := \frac{f(in|g_j)}{f(in|g_{-j})}$, which captures how representative is being an ingroup (in) to group g_j of the receiver relative to group g_{-j} of the non-receiver.

Thus, the DM's *perceived* social proximity of j will be given in the following way:

$$\hat{\gamma}_j = \kappa f(in|g_j)(R(in, g_j, g_{-j}))^{\alpha} \tag{3}$$

where κ is a normalization factor, and $\alpha \geq 0$ is a parameter that reflects to which extent is the DM affected by the bias, where if $\gamma_p = 0$, social context does not distort social proximity. However, if $\gamma_p > 0$, then G_s will distort perceived social proximity. Premise 2 is represented under the following assumption:

P2. $\alpha > 0$.

C Hypotheses

Following the premises P1 and P2 and equations (2) and (3) we can derive the following observations for the effect of social context on the allocation decisions of decision-maker i towards receiver j.

Hypothesis C.1. *If premises P1 and P2 are satisfied, allocation x_j^* from the DM towards a U.S. national receiver in the diverse context is higher than in the homogeneous context.*

Proof. A change from a homogeneous to a diverse context decreases the value of $f(in|g_{-j})$, making being type in more representative for group g_j. Given P2, this generates an increase of perceived social proximity $\hat{\gamma}_j$. Given P1, the increase in perceived social proximity generates an increase in x_j^*. □

Hypothesis C.2. *If premises P1 and P2 are satisfied, allocation x_j^* from the DM towards a foreigner receiver in the diverse context is lower than in the homogeneous context.*

Proof. A change from a homogeneous to a diverse context increases the value of $f(in|g_{-j})$, making being type in less representative for group g_j. Given P2, this generates an decrease of perceived social proximity $\hat{\gamma}_j$. Given P1, the decrease in perceived social proximity generates an decrease in x_j^*. □

D Instructions

Figure 14: Consent.

Welcome to this research project! We very much appreciate your participation.

Procedures
The study consists of a decision that might affect your final payoffs and a series of questions. Please make sure to always read the instructions carefully and answer truthfully.

Participation
Participation in this research study is completely voluntary. You have the right to withdraw at anytime or refuse to participate entirely without jeopardy to future participation in other studies conducted by us.

Confidentiality
Your answers to the questions will be anonymous. That is, your name or any information that could identify you will not be asked or attached to your answers. Data may be shared in anonymized form in open science repositories.

Payment
For completing this survey, you will be automatically enrolled in a lottery by the research team where you can win up to 40 USD (equivalent in points), where 1 in every 100 participants will be selected.

Questions about the Research
If you have questions regarding this study, you may contact: thechoicelab@nhh.no

Consent
Please write 'ACCEPT' in the box below if you have understand the above and wish to participate in this study.

Figure 15: Page 1.

This is a study that will be run on multiple samples of respondents coming from different places.

Before starting the survey, we will match you with two other respondents from the study. The three of you, together, will participate in a lottery to **win a total of $50 as a bonus** (1 in every 100) that will be distributed between the three.

If you win, the three of you will share the $50 bonus. However, **the amount each one receives depends on your choice.** Therefore, please **read the instructions carefully.**

→

Figure 16: Page 2.

You have been matched to the following **two other participants (A and B)**. Below, for each individual you have information about:

1. The **sample** they were drawn from.
2. The **initial distribution** of the **$50 bonus**.

Either A or B will be randomly selected. You will make a decision regarding only the selected participant.

→

Figure 17: Page 3.

Figure 18: Closeness elicitation

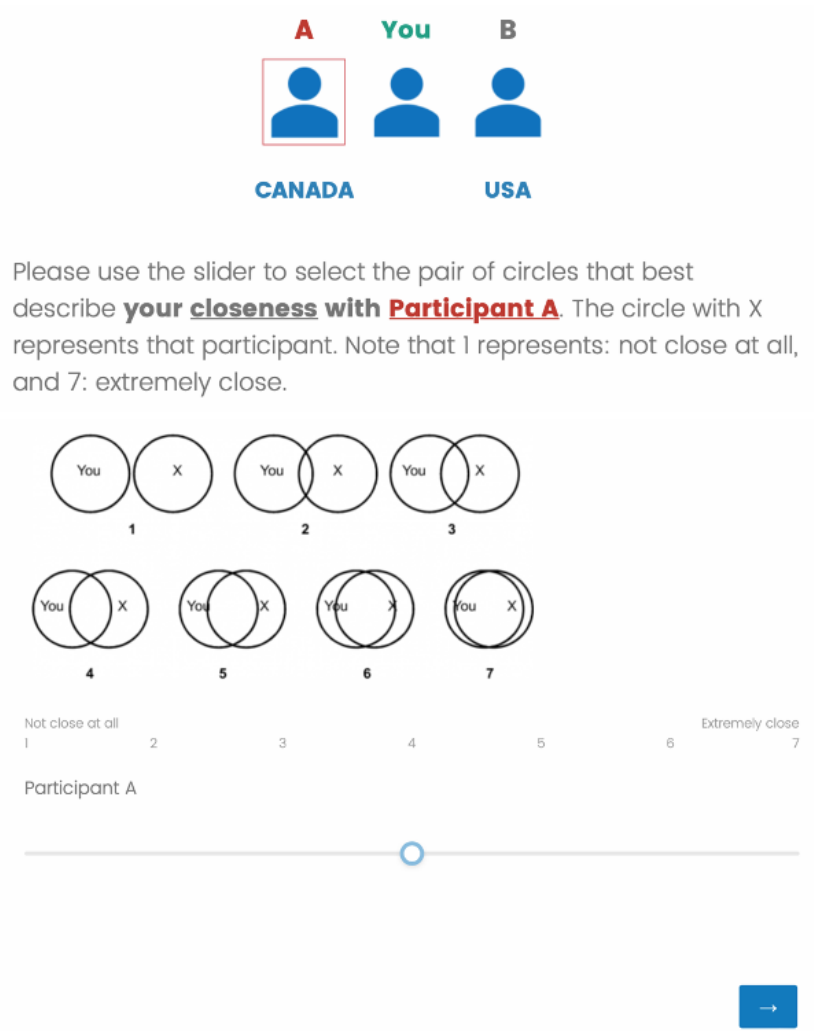


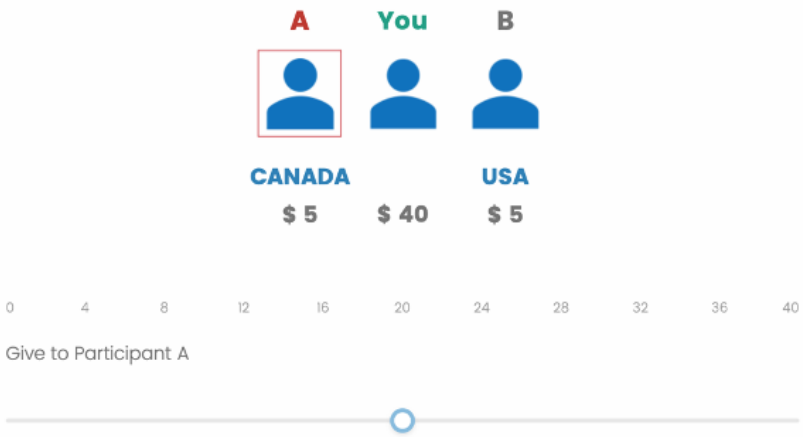
Figure 19: Allocation Decision

You have the possibility to redistribute **between yourself and Participant A**. You can now:

- Give none or up to $40, of the bonus initially allocated to you, to **participant A**.

The final bonus of **you** and **participant A** is determined by your choice. The final bonus of **Participant B**, is $5 independent of what you choose. Neither A or B are able to change their bonuses, and will not know by whom or how the bonuses were determined.

Select in the slider the amount in dollars ($) you decide to reallocate to participant A. The figure below will reflect the choice you make.



Please confirm your choice in the slider. Remember this choice **determines the final bonuses for this survey.**

I confirm the choice in the slider.

→

Figure 20: Attention check

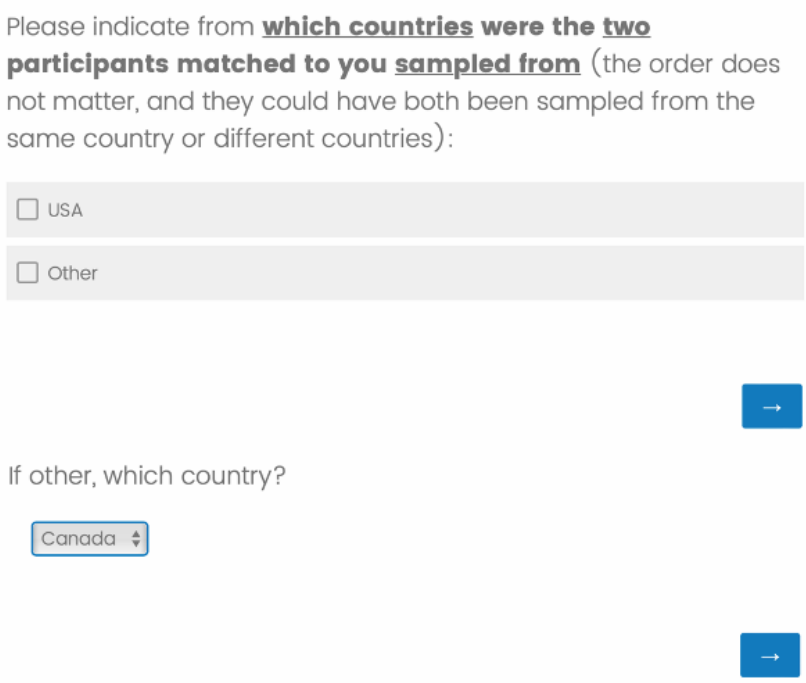


Figure 21: Open-ended question

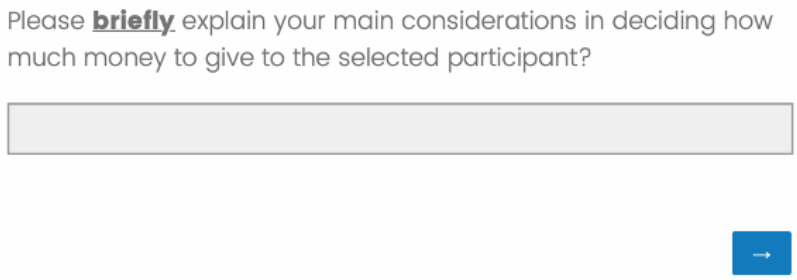


Figure 22: Policy views

You are now going to read a number of statements. In each case, we want you to say whether you Strongly Agree, Somewhat Agree, Do not agree or disagree, Somewhat Disagree, Strongly Disagree. **Read carefully each statement**.

The government should use local tax revenues to fund health insurance only within the local communities they were raised, **rather** than to fund health insurance across all communities nationwide.

- ○ Strongly agree
- ○ Somewhat agree
- ○ Do not agree or disagree
- ○ Somewhat disagree
- ○ Strongly disagree

The government should redistribute local tax revenues as welfare payments across all communities nationwide, **rather** than only within the local communities they were raised.

[options]

The government should send foreign aid to countries that are our international allies, **rather** than to countries that are in most need of help.

[options]

The government should take measures to ensure no child of the world is disadvantaged in access to education, the labor force, and marriage, **rather** than focusing on American children.

- ○ Strongly agree
- ○ Somewhat agree
- ○ Do not agree or disagree
- ○ Somewhat disagree
- ○ Strongly disagree

→

Figure 23: Altruism elicitation

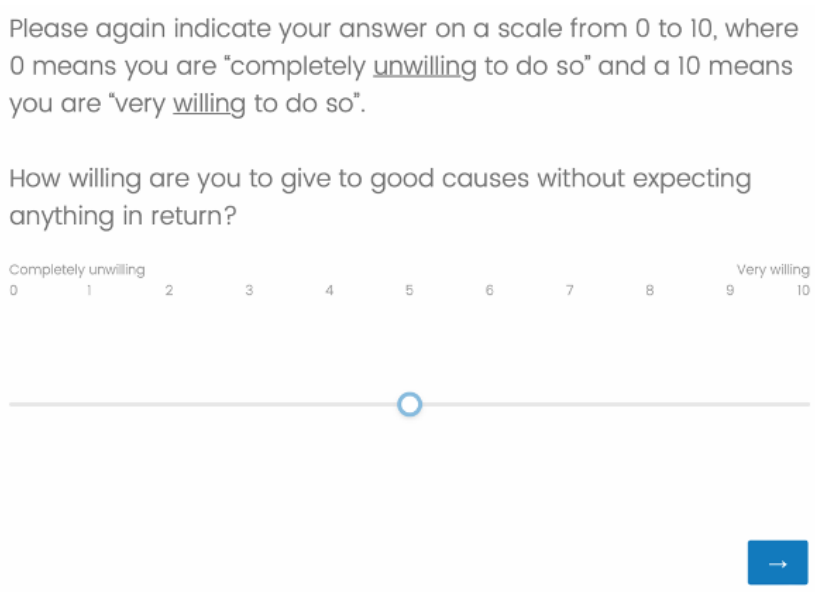


Figure 24: Background demographic information. Part 1.

Background questionnaire

Please answer the following questionnaire. You must answer these questions truthfully.

What is your age?

Which party did you support in the last U.S. elections?

- ○ Republican
- ○ Democrat
- ○ Independent
- ○ I didn't support any party

Generally speaking, do you usually think of yourself as a Republican, a Democrat, an Independent, or something else?

- ○ Democrat
- ○ Independent
- ○ Republican
- ○ Other

What is the highest level of school you have completed or the highest degree you have received?

- ○ High school graduate or less
- ○ Bachelor's degree in college (4-year)
- ○ Master's degree
- ○ Doctoral degree
- ○ Professional degree (JD, MD)

Figure 25: Background demographic information. Part 2.